Bazil-uddin-khan

EX 5.2 Done

Q1). $Ax = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

so eigenvalue $= -1$

Q2) Done

Q3). $Ax = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix} = 2\vec{x}$

Q4) Done.

Q5). a).

$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$

$(1-\lambda)(3-\lambda) - 8 = 0$

$\lambda^2 - 4\lambda + 3 - 8 = 0$

$\lambda^2 - 4\lambda - 5 = 0$

$\lambda^2 - 5\lambda + \lambda - 5$

$\lambda(\lambda-5) + 1(\lambda-5)$

$\lambda = -1, \lambda = 5$

$\lambda = -1$ $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \vec{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} 2 & 4 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $\begin{pmatrix} -2s \\ s \end{pmatrix} \to s \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $x_1 = -2x_2$, $x_2 = s$ $\boxed{(-2,1) \text{ basis for } \lambda = -1}$

$\lambda = 5$ $\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \vec{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\boxed{s\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ basis for } \lambda = 5}$

$\left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $x_1 = x_2$, $x_1 = s$

Q5) b). $\begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} -2-\lambda & -7 \\ 1 & 2-\lambda \end{bmatrix}$

$(-\lambda+2)(-\lambda-2) + 7 = 0$

$\lambda^2 + 2\lambda - 2\lambda + 4 + 7 = 0$

Q5 c, d

Q6)

(a). $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$

$(-\lambda+2)(-\lambda+2)$

$\lambda^2 - 4\lambda + 3 = 0$

$\lambda^2 - 2\lambda - 2\lambda + 3 = 0$

$\lambda(\lambda-2) - 2(\lambda-2) = 0$

$\lambda = 2$, $\lambda_1 = 3$, $\lambda_2 = \lambda$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} x$

$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$x_2 = 0$, $x_1 = 0$, basis

Q7). $\begin{bmatrix} -4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} -4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{bmatrix}$

$(-4-\lambda)\left[(-\lambda+1)(-\lambda+1)\right] + 1\left(0 + 2(1-\lambda)\right)$

$(-\lambda-4)\left[\lambda^2 - 2\lambda + 1\right] + 2 - 2\lambda$

$= -\lambda^3 + 2\lambda^2 - \lambda - 4\lambda^2 + 8\lambda - 4 + 2 - 2\lambda = 0$

$= \lambda^3 + 2\lambda^2 - 6\lambda + 2 = 0$ $\quad \lambda^3 + 2\lambda^2 - 5\lambda + 2 = 0$

$\lambda^3 + 2\lambda^2 - 6\lambda + 2 = 0$

$\lambda^2(\lambda+2)$

$1+2-5+2$

$\begin{array}{c|cccc} 1 & 1 & 2 & -5 & 2 \\ & & 1 & 3 & -2 \\ \hline & 1 & 3 & -2 & 0 \end{array}$

$(\lambda-1)\underbrace{(\lambda^2 + 3\lambda + 2)}_{} = 0$

$\lambda = 1$,

Q7).

$\begin{bmatrix} -4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} -4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{bmatrix}$

$(-4-\lambda)\left[(-\lambda+1)(-\lambda+1)\right] + 1(-2+\lambda)$

$(-\lambda-4)\left[\lambda^2 - 2\lambda + 1\right] - 2 + \lambda = 0$

$\begin{array}{c|cccc} -2 & 1 & -1 & -16 & -20 \\ & & -2 & 6 & 20 \\ \hline & 1 & -3 & -10 & 0 \end{array}$

Q9) $\begin{bmatrix} 6 & 3 & -8 \\ 0 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 6-\lambda & 3 & -8 \\ 0 & -2-\lambda & 0 \\ 1 & 0 & -3-\lambda \end{bmatrix}$

$= \begin{bmatrix} -\lambda+6 & 3 & -8 \\ 0 & -\lambda-2 & 0 \\ 1 & 0 & -\lambda-3 \end{bmatrix}$

$\lambda^2 - 3\lambda - 10 = 0$

$\lambda^2 - 5\lambda + 2\lambda - 10 = 0$

$\lambda(\lambda-5) + 2(\lambda-5)$

$= (-\lambda+6)\left[(-\lambda-2)(-\lambda-3)\right] - 3\left[0\right] - 8\left[\lambda+2\right] = 0$

$= (\lambda+6)\left[\lambda^2 + 5\lambda + 6\right] - 8\lambda - 16 = 0$

$= (-\lambda^3 - 5\lambda^2 - 6\lambda + 6\lambda^2 + 30\lambda + 36) - 8\lambda - 16 = 0$

$= -\lambda^3 - 5\lambda^2 + 6\lambda^2 - 6\lambda + 30\lambda - 8\lambda + 36 - 16 = 0$

$= -\lambda^3 + \lambda^2 + 16\lambda + 20 = 0$

$\lambda^3 - \lambda^2 - 16\lambda - 20 = 0$

$(\lambda+2)(\lambda^2 - 3\lambda - 10) = 0$

$= -8 - 4 + 32 - 20$

$\lambda = -2, \lambda = 5, \lambda = -2$

Q15). $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix} \to (-\lambda+1)(-\lambda+3) - 8 = 0$

$\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}$ $\quad \lambda^2 - 4\lambda - 5 = 0$

$\lambda^2 - 5\lambda + \lambda - 5 = 0$

$(\lambda-5)^2 = 0$

$\lambda = 5, \lambda = -1$

$\left[\begin{array}{cc|c} -4 & 4 & 0 \\ 2 & -2 & 0 \end{array}\right] \to R_1 + 2R_2$

$\left[\begin{array}{cc|c} -4 & 4 & 0 \\ 0 & 0 & 0 \end{array}\right] \to -x_1 + x_2 = 0$ $\quad \lambda = 5$ basis

$x_1 = x_2$

$x_2 = s$ $\; s\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

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